Unit: 6 Unit 2: Exponents Radicals
Topic: Exponential Equations

Name: ________________
Date: ____________

Advanced (Carolina Reaper)

Q1. Solve for x in $2^x = 32$
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = 7$

Q2. If $3^x = 81$, what is x?
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q3. Solve $4^x = 256$ for x
(A) $x = 2$
(B) $x = 3$
(C) $x = 4$
(D) $x = 5$
(E) $x = 6$

Q4. A bookshelf has 5 shelves, and each shelf can hold 2^x books. If the bookshelf is currently holding 160 books, what is x?
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = 7$

Q5. A bakery sells 3^x loaves of bread per day. If they sold 243 loaves yesterday, what is x?
(A) $x = 4$
(B) $x = 5$
(C) $x = 6$
(D) $x = 7$
(E) $x = 8$

Q6. Solve for x in $2^x + 2^x = 96$
(A) $x = 4$
(B) $x = 5$
(C) $x = 6$
(D) $x = 7$
(E) $x = 8$

Q7. If $x^2 = 2^{10}$, what is x?
(A) $x = 2^3$
(B) $x = 2^4$
(C) $x = 2^5$
(D) $x = 2^6$
(E) $x = 2^7$

Q8. Solve $4^x - 2^x = 60$ for x
(A) $x = 3$
(B) $x = 4$
(C) $x = 5$
(D) $x = 6$
(E) $x = 7$

Open Ended Questions (FRQ)

Q9. A person invests

$$2,000$$

in a savings account with an annual interest rate of 5%. If the interest is compounded annually, how much will the person have in the account after x years? Write an equation to represent the situation and solve for x when the total amount is

$$3,000$$

. Required work and calculations should be shown.

Q10. A population of bacteria grows according to the equation $P = 2^x$, where P is the population size and x is the number of hours. If the initial population size is 100, how many hours will it take for the population to reach 1,600? Show all work and calculations. Use the equation to solve for x.

Chef's Tips

- Tip 1: Emphasize the importance of showing work and calculations for free-response questions.
- Tip 2: Make sure students understand the context of each problem and how it relates to exponential equations.
- Tip 3: Encourage students to check their work and calculations carefully to avoid mistakes.